

HOUR: 1

1. Table below shows the selected elements and their proton numbers:

Elements	Proton Number
Al	13
P	15
Cl	17
Fe	26
Zn	30

- a) Write the Lewis dot symbol for **Al**, **P** and **Cl**.
 b) Write the electronic configuration for the ions formed by **ALL** elements.
 c) State the type of stability of the ions formed from these elements. (CLO3, C3)
2. How would an atom acquire octet configuration when it forms bond with another atom?
PSPM JAN 2000 (CLO1, C2)
3. a) How is an ionic bond formed?
 b) By using Lewis dot symbol, show the formation of:
 i) MgBr_2
 ii) Al_2O_3
 iii) BaO (CLO3, C3)
4. a) How is a covalent bond formed?
 b) Use Lewis dot symbols to show the sharing of electrons between hydrogen and fluorine atoms. Label the electron pairs as bonding pair(s) and lone pair(s). (CLO3, C3)
5. Describe the formation of dative bond by using NH_4^+ , H_3O^+ , Al_2Cl_6 , and BF_3NH_3 , as examples.
(CLO3, C3)
6. The **TABLE 1** below shows two elements with their respective proton numbers.

Elements	U	V
Proton Number	20	9

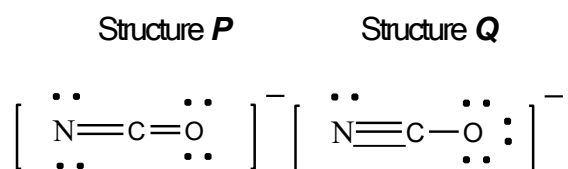
TABLE 1

- Based on the table above, state the type of bond formed between element **U** and **V**. Explain how the bond is formed.
PSPM 2000 (CLO3, C3)
7. Elements **X** and **Y** have proton number of 12 and 7, respectively. Write the chemical formula of the compound formed between **X** and **Y**. Show the formation of the compound using Lewis dot symbol.
(CLO3, C4)

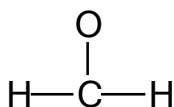
HOUR: 2

8. Draw the simplest Lewis structure for each of the following molecules. Determine whether these molecules satisfy the octet rule. (CLO3, C4)
- NCl_3
 - ClF_3
 - BeCl_2
 - H_2S
 - AlBr_3
 - SeF_4
 - NO
 - CN^-
 - NO^+
 - O_3

9. Two resonance structures for NCO^- are as follows:



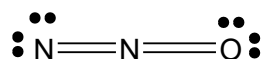
- Determine the formal charge of each atom in both structures **P** and **Q**.
 - Determine the most plausible structure. Explain. (CLO3, C4)
10. Formaldehyde (CH_2O), a liquid with an unpleasant odour with the skeletal structure



Draw the Lewis structure for formaldehyde and determine the formal charge of each atom.

(CLO3, C3)

11. Draw all possible Lewis structures for POCl_3 . Select the best Lewis structure. (CLO3, C3)
12.
 - Define resonance structures.
 - Draw the resonance structures for NO_3^- and SO_3 . (CLO3, C4)
13. Dinitrogen oxide molecule, N_2O , has three resonance structures. The atomic arrangement in the molecule is NNO . One of the resonance structures is shown below:



- Draw the other two resonance structures.
- Calculate the formal charge of each atom in the resonance structures in 13. a)
- Determine the most plausible Lewis structure between the two resonance structures. Give your reasons. (CLO3, C4)

MID SEM 2012 / 2013 (CLO3, C4)

HOUR: 3

14. Using VSEPR model, predict the molecular geometries of:
 - a) O_3
 - b) PBr_5
 - c) NO_2^+
 - d) $SnCl_3^-$(CLO3, C3)
15. Predict the shape of SiF_4 , SF_4 and XeF_4 molecules. Explain why the shapes differ. (CLO3, C4)
16. Antimony pentafluoride, SbF_5 , reacts with XeF_4 and XeF_6 to form the ions XeF_3^+ , SbF_6^- and XeF_5^+ . Describe the geometries of these ions. (CLO3, C3)
17. The nitronium ion, NO_2^+ is linear in shape but the nitrite ion NO_2^- is not linear. Explain the observation. (CLO3, C3)
18. Iodine combines with chlorine to form ICl_2^+ and ICl_2^- ions.
 - a) Draw the Lewis structures and predict the shapes of both ions based on the Valence Shell Electron-Pair Repulsion (VSEPR) theory.
 - b) Determine whether both the ions are polar or non polar. PSPM 2001 / 2002 (CLO3, C4)

HOUR: 4

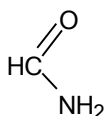
19. For each of the molecules given; PCl_3 , PCl_5 and $POCl_3$,
 - a) Draw a Lewis structure and name the shape based on molecular geometry.
 - b) Predict the bond angle and deduce the polarity. PSPM 2012/2013 (CLO3, C4)
20.
 - a) Define dipole moment.
 - b) How can a molecule be non-polar although it has polar bonds?
 - c) Explain whether each of the following molecules is polar or non-polar. (CLO3, C4)
 - i) SO_2
 - ii) HBr
 - iii) BF_3
 - iv) SF_6
 - v) NH_3
 - vi) H_2Se
21. An atom X has 5 valence electrons. X reacts with fluorine gas to form XF_3 and XF_5 compounds. For each compound,
 - a) draw the Lewis structure,
 - b) predict the electron pair geometry and molecular geometry and draw the molecular geometry and state the bond angle(s). PSPM 2015 / 2016 (CLO3, C4)
22. Xenon, Xe which is located in the noble gas group attains octet configuration. Xe forms compounds such as the fluoride compounds, XeF_2 and XeF_4 .
 - a) State the number of bonding pairs and lone pairs surrounding the central atom Xe in XeF_2 and XeF_4 . How does Xe atom accommodate the total number of electrons.
 - b) Give the shapes of XeF_2 and XeF_4 molecules.
 - c) Besides XeF_2 and XeF_4 , give a molecular formula of another fluoride of Xe . PSPM JAN 2000 (CLO3, C3)

HOUR: 5

23. a) Describe the formation of sigma and pi bonds. (CLO1, C2)
 b) By using valence bond theory, explain the bonding in HCl and show the bond formation by the overlapping of atomic orbitals. (CLO3, C3)
24. a) What is meant by hybridisation? (CLO1, C1)
 b) Determine the hybridisation of the underlined atom in each of the following compounds.
 i) $\underline{\text{B}}\text{F}_3$
 ii) $\underline{\text{C}}\text{Cl}_4$
 iii) $\text{CH}_3\underline{\text{C}}\text{H}=\text{CH}_2$ (CLO3, C3)
25. Describe the hybridisation process and draw the overlapping of orbitals for the formation of bonds in each of the following molecules.
 a) CH_3F
 b) AsCl_5
 c) XeF_4
 d) ICl_3
 e) BrF_5
 f) SeF_4
 g) HCN
 h) OCN^-
 i) PH_4^+
 j) ICl_2^- (CLO3, C4)

HOUR: 6

26. Formamide, *I*, is a clear liquid which is miscible with water and is normally used as a chemical feedstock in pharmaceutical industries.



Formamide, *I*

Draw the overlapping of hybrid orbitals for the bonding in *I*. Label all the σ and π bonds.

PSPM DK015 2014 / 2015 (CLO3, C4)

HOUR: 7

27. a) What are dipole-dipole interaction? (CLO3, C3)
 b) Explain how London forces arise between non-polar molecules. (CLO1, C2)
 c) Give the factors that influence the strength of van der Waals forces.
28. Name the type of intermolecular forces that exist between molecules for each of the following species:
 a) I_2
 b) SO_2
 c) CH_4

- d) H_2S
 e) C_6H_6
 f) H_2O (CLO3, C3)

29. Which molecule has a higher boiling point? Explain.

- a) Br_2 or ICl
 b) CH_4 or SiH_4
 c) HCl or Cl_2 (CLO3, C3)

30. a) What is meant by hydrogen bond? (CLO1, C1)
 b) Which has a higher boiling point, ammonia, NH_3 or phosphine, PH_3 ? (CLO3, C3)
 c) Arrange the following compounds in ascending order of boiling point. Explain. (CLO3, C3)
 $\text{C}_2\text{H}_5\text{OH}$ CH_3OCH_3 CH_3COOH

HOUR: 8

31. The boiling points and solubility of ammonia, hydrogen chloride and fluorine are given below:

Compound	Boiling point / $^{\circ}\text{C}$	Solubility in water / mol dm^3 at 298 K
hydrogen chloride	-85	23
fluorine	-188	0
ammonia	-33	18

- a) Explain in terms of intermolecular forces the difference in boiling points between
 i) hydrogen chloride and fluorine.
 ii) ammonia and hydrogen chloride (CLO3, C3)
 b) Explain why the solubility of ammonia and hydrogen chloride are so much higher than that of fluorine. (CLO3, C3)

32. Explain why ice floats on water. (CLO3, C3)

33. Explain the following observation:

Methane, CH_4 (16 g mol^{-1}) has lower boiling point than propane, C_3H_8 (44 g mol^{-1}) whereas water, H_2O (18 g mol^{-1}) has higher boiling point than hydrogen sulphide, H_2S (34 g mol^{-1}).

PSPM 2015 / 2016 (CLO3, C3)

HOUR: 9

34. a) Describe the metallic bond in aluminium by using electron sea model. (CLO3, C3)
 b) Explain why the boiling point of calcium is higher than that of potassium. (CLO3, C3)

35. Using Band Theory, explain the electrical conductivity of Mg, Si and S. (CLO3, C3)

36. Arrange in order of increasing melting points for the solid crystals of copper, Cu, iodine, I_2 and diamond, C. Explain your answer with reference to the attractive forces between the atoms. Which of the solid crystals can conduct electricity in the solid state? Explain.